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Numerical Assessment of Thermal Energy Storage in a Triplex-Tube Heat Exchanger With Al_2O_3-Enhanced Phase Change Material and Tree-Shaped Fins

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Abstract: Thermal energy storage (TES) performance in a triplex-tube heat exchanger incorporating phase change material (PCM) was numerically investigated through the combined use of Al_2O_3 nanoparticles and tree-shaped fins. The proposed configuration was developed to address the intrinsically low thermal conductivity of paraffin-based PCM and to accelerate heat transfer during both charging and discharging processes. The numerical model was validated against previously reported data, with satisfactory agreement obtained between the predicted and reference results. To improve computational efficiency while maintaining numerical accuracy, the spatial mesh resolution and temporal step size were systematically optimised. Transient melting and solidification were modelled using the enthalpy–porosity method, while the thermophysical properties of the nanoparticle-enhanced phase change material (NEPCM) were represented using a homogeneous model. The thermal storage characteristics of the proposed configurations were evaluated in terms of charging and discharging times, temperature distributions, and phase-change evolution. Relative to the corresponding nanoparticle-free configurations, the incorporation of Al_2O_3 nanoparticles reduced the charging time by 8.55% and 9.6% for Cases 1 and 2, respectively. At a nanoparticle concentration of 3%, the introduction of tree-shaped fins reduced the charging time by 44.09% compared with the unfinned configuration. During discharging, reductions of 7.94% and 9.69% were obtained for Cases 1 and 2, respectively, following the incorporation of nanoparticles, while the tree-shaped fin configuration reduced the discharging time by 27.6% relative to the unfinned case. Overall, both nanoparticle incorporation and fin-induced geometric modification were found to enhance TES performance, with the tree-shaped fins producing the substantially greater contribution to the acceleration of both melting and solidification. These findings demonstrate the potential of combining conductive nanoparticles with biomimetic fin architectures for improving the thermal response of PCM-based TES systems.

Keywords: Triplex-tube heat exchanger; Phase change material; Nanoparticle-enhanced phase change material; Al_2O_3 nanoparticles; Tree-shaped fins; Thermal energy storage; Melting and solidification; Enthalpy–porosity method

1 Introduction

The socioeconomic stability of nations across the world has been adversely affected by global challenges, including the depletion of natural resources and the deterioration of environmental conditions. These interconnected problems have increased the need for sustainable development approaches and brought attention to the need for long-term policy implementation together with systematic energy planning. Consequently, developing efficient energy solutions has emerged as a crucial goal for attaining energy security, environmental sustainability, and economic resilience. In this regard, reducing greenhouse gas emissions and preventing environmental degradation depend heavily on the effective utilization of renewable energy resources [1]. The transition to cleaner energy systems is especially important in light of global carbon peaking and carbon neutrality goals, which call for significant

increases in energy efficiency and the reorganisation of conventional energy systems [2]. Due to its extensive availability, sustainability, and little environmental effect, solar energy is regarded as an exceptionally plentiful and promising renewable energy source. Nevertheless, despite its enormous potential, solar energy intrinsically suffers from intermittency and temporal fluctuations brought on by weather conditions, seasonal variations, and day-night cycles. These variations often lead to inconsistencies between energy generation and actual demand requirements, thus limiting the dependability and stability of solar energy systems [3]. To overcome these limitations, the creation of efficient energy storage devices has emerged as a crucial area of study in contemporary energy engineering. In particular, storing renewable energy in appropriate and stable forms is crucial for maintaining a consistent energy supply and enhancing the flexibility of the system [4]. Thermal energy storage (TES) technologies have emerged as extremely promising methods for improving the integration of renewable energy sources and increasing operational stability of sustainable energy systems, especially latent heat systems that use phase change material (PCM).

Buildings may store thermal energy during periods of excess and release it during peak load periods through TES systems, which effectively separate energy supply from end use-demand. This operational flexibility increases the overall energy management by enhancing distribution efficiency, stabilizing system performance, and significantly decreasing peak demand loads [5]. Due to their superior thermophysical and energy storage properties, PCM-based latent heat thermal energy storage (LHTES) systems are particularly attractive for building-related applications. PCMs can absorb or release substantial quantities of latent heat during phase transitions, and their energy storage densities are around five to fourteen times greater than those of traditional sensible heat storage materials [6]. They are also very appropriate for building heating and cooling applications because of their near-isothermal charging and discharging behavior, which improves thermal control, load-shifting capabilities, and overall energy efficiency [7]. The most recent developments and limitations in PCMs for solar energy applications were examined by Awan et al. [8]. A thorough analysis of important methods for improving PCM charging and discharging efficiency in triplex-tube heat exchanger was carried out by Leong et al. [9]. Fin installation methods, the geometry of the triplex tube heat exchanger, the use of metal foams, nano-enhanced PCM, and the temperature of the heat transfer fluid (HTF) are among the crucial augmentation strategies. Fin architectures can improve the thermal performance of latent energy storage devices, according to a review study by Zhang et al. [10]. They demonstrated that novel helix-shaped fin designs and fin topology optimisation provide significant research potential due to their superior thermal performance when compared to conventional fin configurations. Gasia et al. [11] investigated the novel idea of dynamic melting within a heat exchanger to improve the melting process. A low-viscosity potassium water solution serves as the HTF to lower pumping energy consumption, while water serves as the PCM. The PCM is located in the annular area of the heat exchanger, which is made up of two concentric vertical pipes. Using an external force to recirculate the liquefied PCM increases the overall rate of heat transmission. Rokhforouz and Sheikholeslami [12] used PCM to study nanoparticles and fins for increasing the efficiency of a parabolic solar collector. The study revealed that fins and nanoparticles can improve the efficiency of melting and solidification operations. In a novel heat exchanger with leaf-shaped fins, Ji et al. [13] examined the melting performance of molten salt phase transition material enhanced by TiO_2 and Al_2O_3 nanoparticles. Adding high thermal conductivity fins to PCMs is one improvement method that increases the heat exchange surface area and can optimise the melting process in latent heat TES systems. Extensive investigations have explored the possible advantages of integrating fins in different configurations in heat exchanger systems. In order to improve the charging process of a single-PCM (RT82)-filled storage system, Al-Abidi et al. [14] initially investigated the use of rectangular fins on both the inner and outer surfaces of a latent thermal storage system. According to their findings, the melting rate was significantly increased by adding internal and exterior fins. The thermal characteristics of a TES system with bifurcated fractal leaf-vein-shaped fins were studied by Jiang et al. [15], emphasising its function in accelerating the melting process of lauric acid. Using a numerical model, they evaluated the effects of fin length, number, and placement on heat transfer performance. Their results demonstrated that increasing the length ratio and fin height accelerates the charging process.

According to available literature, limited research has focused on the innovative design of energy storage systems using nanomaterials, tree-shaped fins, and hexagonal tubes. In the present study, a novel hexagonal structure with tree-shaped fins for heat storage was investigated to improve system performance. The middle tube was filled with RT42, a paraffin-based PCM. To improve the effective heat conductivity, Al_2O_3 nanoparticles were used. The outlet of the evacuated solar system provides the HTF that circulates through the inner and outer pipes. Consequently, a high temperature range is considered a boundary condition for the inner and outer surfaces. To increase the rates of melting and solidification, two strategies were employed: (1) the installation of tree-shaped fins and (2) the dispersion of Al_2O_3 nanoparticles in paraffin (RT42). To produce two different configurations with an identical volume of PCM, one geometric parameter was treated as a variable, while the other parameter was calculated to satisfy the constraint of maintaining equal PCM volume. The laminar two-dimensional charging and discharging operations were modeled using the finite volume method. An optimized number of elements was used to reduce the computational cost, and an implicit technique with an unsteady term was employed with an optimized time step. The numerical model was verified using previously published data, and good agreement was obtained. Plots and

contours were used to describe the thermal field variations over time in the outputs.

2 Configuration of the Thermal Storage Unit and Tree-Shaped Fin Geometry

2.1 Proposed Physical Model Description

A TES system is combined with an evacuated tube solar collector, which supplies the required domestic hot water. As shown in Figure 1, a triplex-tube heat exchanger with tree-shaped fins is suggested. The initial temperature of the PCM was set to 300 K, while the inner and outer heat transfer walls were maintained at a constant temperature of 323 K during the charging process. The PCM was initially at 323 K, while the inner and outer heat transfer walls were maintained at a constant temperature of 300 K during the discharging process. The HTF's thermal impact was represented by the specified isothermal wall boundary conditions rather than being expressly included as a distinct computing domain. The PCM volume was maintained identical to ensure a fair comparison between the two cases.

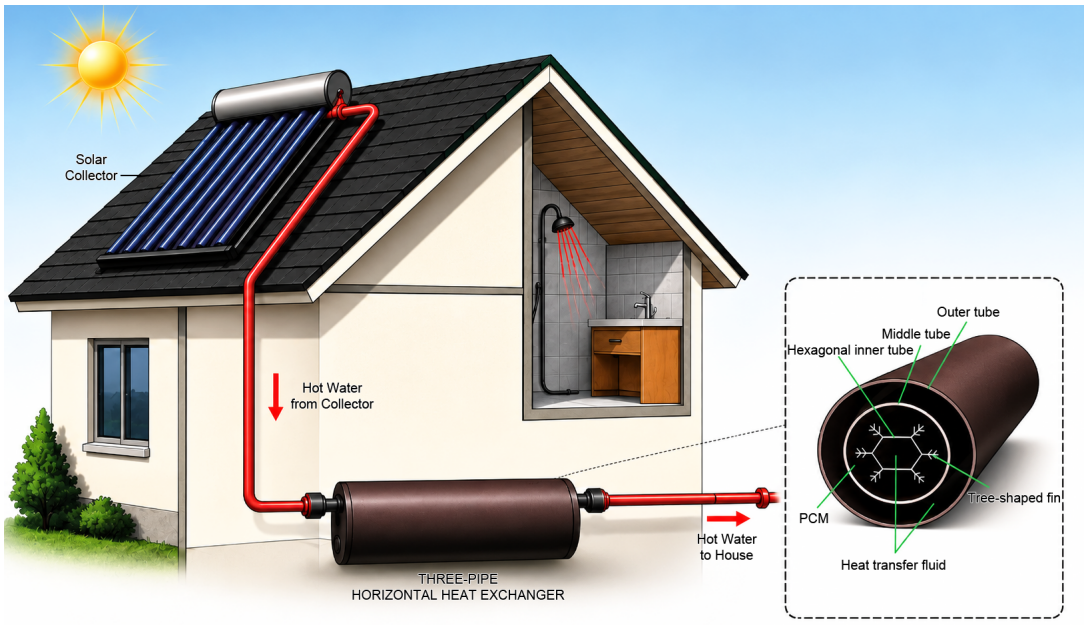


Figure 1. Geometrical configuration of the TES system incorporating tree-shaped fins

Note: PCM = phase change material; TES = thermal energy storage.

The arrangement without tree-shaped fins is shown in Case 1, whereas the inner hexagonal tube is equipped with tree-shaped fins in Case 2. In both configurations, the side length of the inner regular hexagonal tube was 10 mm. Since the tree-shaped fins occupy a portion of the PCM domain, the outer diameter of Case 1 was 58.3 mm to account for the volume occupied by the fins and preserve the same PCM volume in both cases. The outer diameter of Case 2 was kept at 60 mm. Accordingly, in Case 1 and Case 2, the radial distance between the hexagonal tube's outermost vertex and the outer wall was 19.15 mm and 20 mm, respectively. In both configurations, the resulting PCM cross-sectional area was 2409.67 mm^2, while the total cross-sectional area occupied by the tree-shaped fins in Case 2 was 157.95 mm^2. The difference between the outer circular cross-sectional areas of Cases 2 and 1 was 157.95 mm^2, corresponding to the total cross-sectional area occupied by the tree-shaped fins in Case 2. Since the same axial length was considered for both configurations, the identical PCM cross-sectional areas ensured identical PCM volumes. As a result, Cases 1 and 2 were compared under an equal-PCM-volume condition. The corresponding geometrical dimensions of the hexagonal inner tube and the tree-shaped fins are shown in Figure 2. Paraffin (RT42) was used to fill the central annulus, and Al_2O_3 nanoparticles were added to improve the characteristics. Two distinct nanoparticle volume fractions were examined: $\phi = 0$ for pure RT42 PCM and $\phi = 0.03$ for the nanoparticle-enhanced phase change material (NEPCM) containing 3 vol.% Al_2O_3. Both nanoparticle concentrations were evaluated for Case 1 and Case 2. The homogeneous mixture model was used to assess the thermophysical characteristics of the nanoparticle-enhanced phase change material for the nanoparticle-enhanced configurations. The thermophysical characteristics of the materials used in this study are summarised in Table 1 [16, 17]. The impact of gravity has been included in the velocity equation, and a transient melting source term has been added. Furthermore, the governing energy equation has been discretised and solved using the finite volume technique. Only half of the enclosure was examined because of the symmetric geometry, and paraffin melting was investigated using a two-dimensional computational domain. The PCM was characterised as a Newtonian medium with little compressibility, while

buoyancy-induced density changes were represented using the Boussinesq approximation. Volumetric expansion and viscous dissipation were ignored in the present model. The following governing equations have been used to describe the laminar unsteady phenomena [18].

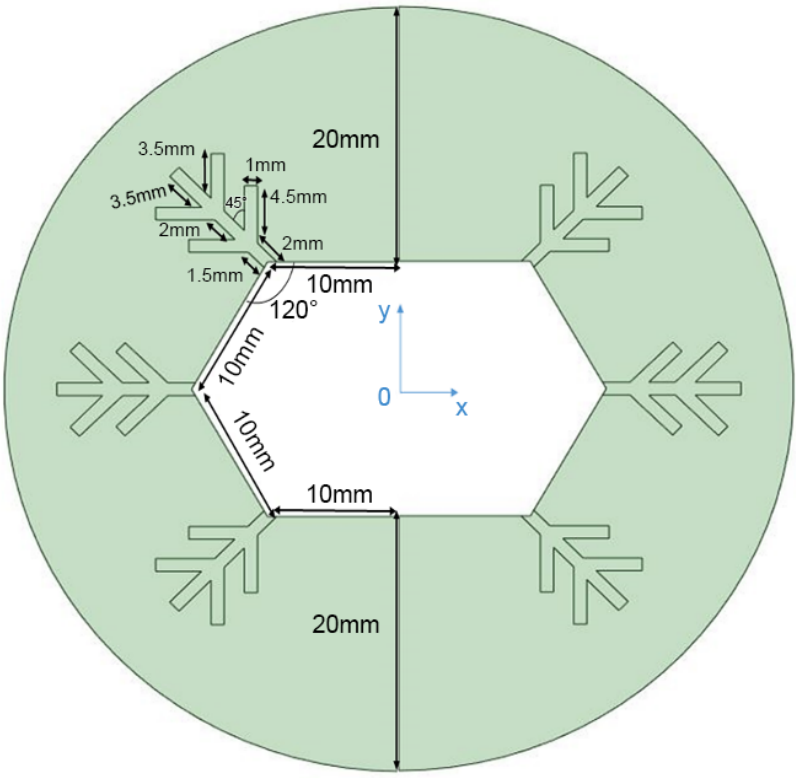


Figure 2. Schematic representation of the heat exchanger geometry adopted in the present numerical model

Table 1. Material properties used in this study [16, 17]

Physical Property	Al_2O_3	RT42	Unit
ρ	3,600	760	kg m^{-3}
μ	—	0.02351	kg m^{-1} s^{-1}
β	—	0.0005	K^{-1}
C_p	765	2,000	J kg^{-1} K^{-1}
k	36	0.2	W m^{-1} K^{-1}
Melting point	—	311.15–315.15	K
Latent heat	—	165	kJ kg^{-1}

Note: "—" indicates that the property is not applicable or the data are not provided.

Continuity equation:

$$\nabla \cdot \vec{V} = 0 \tag{1}$$

Momentum equation:

$$\frac{\partial \vec{V}}{\partial t} + \vec{V} \cdot \nabla \vec{V} = -\frac{(1-\gamma)^2}{\gamma^3 + 0.001} A_{mush} \vec{V} + \frac{1}{\rho} \left[-\nabla p + \mu \nabla^2 \vec{V} + \rho \beta \vec{g} \left(T - T_{ref} \right) \right. \tag{2}$$

Energy equation:

$$\frac{\partial \left(\rho H_{tot} \right)}{\partial t} + \nabla \cdot \left(\rho \vec{V} H_{tot} \right) = \nabla \cdot \left(k \nabla T \right) \tag{3}$$

where, H_{tot}, P, V, γ, A_{mush}, and T represent total enthalpy, pressure, velocity field, liquid-phase fraction, mushy-zone coefficient, and temperature, respectively. The absolute viscosity, density, thermal conductivity, and

thermal expansion coefficient are represented by the parameters μ, ρ, k, and β, respectively, while the gravitational acceleration is indicated by g. The buoyancy term in the momentum equation was evaluated using the Boussinesq approximation with a reference temperature (T_{ref}) of 300 K. A mushy zone parameter of 10^5 was employed in the present numerical study.

The PCM enthalpy comprises both sensible and latent heat contributions and can be formulated as follows [19]:

$$H_{tot} = H_{sens} + H_{lat} \tag{4}$$

$$H_{sens} = H_{ref} + \int_{T_{ref}}^{T} C_p dT = H_{ref} + C_p \int_{T_{ref}}^{T} dT \tag{5}$$

The specific heat value is C_p, while the latent and sensible heats are H_{lat} and H_{sens}, respectively. The formulation for the latent heat component is as follows [20]:

$$H_{lat} = \Gamma\gamma \tag{6}$$

where, Γ is the melting latent heat and γ is the liquid-phase fraction. The liquid fraction, γ, varied with temperature within the phase-change interval according to Eq. (7), whereas the thermal conductivity, k , was assumed to be constant throughout the phase-transition process. The liquid phase is described as follows [21]:

$$\gamma = \begin{cases} \frac{T-T_{sol}}{T_{liq}-T_{sol}} & \text{if} \quad T_{sol} < T < T_{liq} \\ 0 & \text{if} \quad T < T_{sol} \\ 1 & \text{if} \quad T > T_{liq} \end{cases} \tag{7}$$

The terms “sol” and “liq” refer to the solid and liquid phases of the PCM, respectively. Natural-convection effects were taken into consideration using the Boussinesq approximation. With the exception of the buoyancy term of the momentum equation, where its temperature dependence was represented by the linearized Boussinesq relation provided in Eq. (8), density was assumed to remain constant in all governing equations, which is expressed as follows [22]:

$$\rho = \rho_{nef}\left[1 - \beta\left(T - T_{nef}\right)\right] \tag{8}$$

where, ρ_{ref} is the reference density of the PCM, while β is the thermal expansion coefficient, and T_{ref} = 300 K is the reference temperature. T_{sol} and T_{liq} stand for the temperatures that correspond to the PCM’s solidus and liquidus limits, respectively.

The thermophysical properties of the NEPCM are computed using the following formulas [23, 24]:

$$\rho_{NEPCM} = \phi\rho_p + (1-\phi)\rho_{PCM} \tag{9}$$

$$(\rho C_P)_{NEPCM} = \phi\left(\rho C_P\right)_P + (1-\phi)\left(\rho C_P\right)_{PCM} \tag{10}$$

$$k_{NEPCM} = \left[\frac{k_p + 2k_{PCM} + 2\phi\left(k_p - k_{PCM}\right)}{k_p + 2k_{PCM} - \phi\left(k_p - k_{PCM}\right)}\right] k_{PCM} \tag{11}$$

$$(\rho L)_{NEPCM} = (1-\phi)(\rho L)_{PCM} \tag{12}$$

$$\mu_{NEPCM} = \frac{\mu_{PCM}}{(1-\phi)^{2.5}} \tag{13}$$

The nanoparticle concentration is represented by ϕ in the aforementioned equations, while the subscript p denotes solid particles.

2.2 Details of the Numerical Method

The governing equations in this study were numerically solved with ANSYS Fluent, a well-known computational fluid dynamics software application. Two-dimensional transient flow was simulated. The study used a pressure-based unsteady solver to examine the system and focused on laminar flow. The PREssure STaggering Option (PRESTO) approach was used for pressure discretisation, while the Semi-Implicit Method for Pressure-Linked Equations (SIMPLE) algorithm was used for pressure–velocity coupling. Further scalars were discretised using

the 2nd-order upwind method. To arrive at a converged solution, the residual values must be smaller than 10-5. The transient numerical results produced with ANSYS Fluent were used to directly calculate the charging and discharging times. The temporal simulation results were used to obtain the phase-change process completion time for each configuration under investigation. The final charging time was determined by the end of the melting phase, whereas the final discharging time was determined by the end of the solidification process. Quantitative comparisons between the various configurations were then based on these completion times. The relative difference between the comparable completion times was used to calculate the percentage reduction in each case.

3 Results and Discussion

Environmental degradation and energy scarcity are the two key factors that might affect global economic development. In recent years, thermal storage technologies have been used alongside renewable energy sources to maintain the balance between energy demand and supply. Evacuated tube systems may be integrated with TES units to provide a stable domestic hot-water supply. In this work, unique tree-shaped fins were introduced to enhance the performance of these units. The same amount of paraffin was considered in two configurations. PCM was located in the center of the heat exchanger, and the heated fluid passes via the inner and outer tubes. The temperatures of the outer and inner walls are sustained at 323 K, which aligns with the temperature of an evacuated solar system that captures sunlight and generates domestic hot water. The domain was set to an initial temperature of 300 K, and the paraffin was encouraged to melt by the thermal boundary conditions placed on the inner and outer walls. RT42 was chosen as the PCM due to the operating temperatures of the outer and inner surfaces, and Al_2O_3 nanoparticles were added to improve performance. A homogeneous model can be used since the maximum volume fraction of Al_2O_3 nanoparticles is 0.03. In a two-dimensional domain, where buoyant forces have a major impact on the transient heat transfer process, the behaviour of PCM during charging and discharging processes has been studied. By adding related source terms to the temperature and velocity equations, laminar flow formulation has been established. The finite volume technique has been used to find solutions for various models. In numerical investigations, the results must be verified to be independent of mesh resolution.

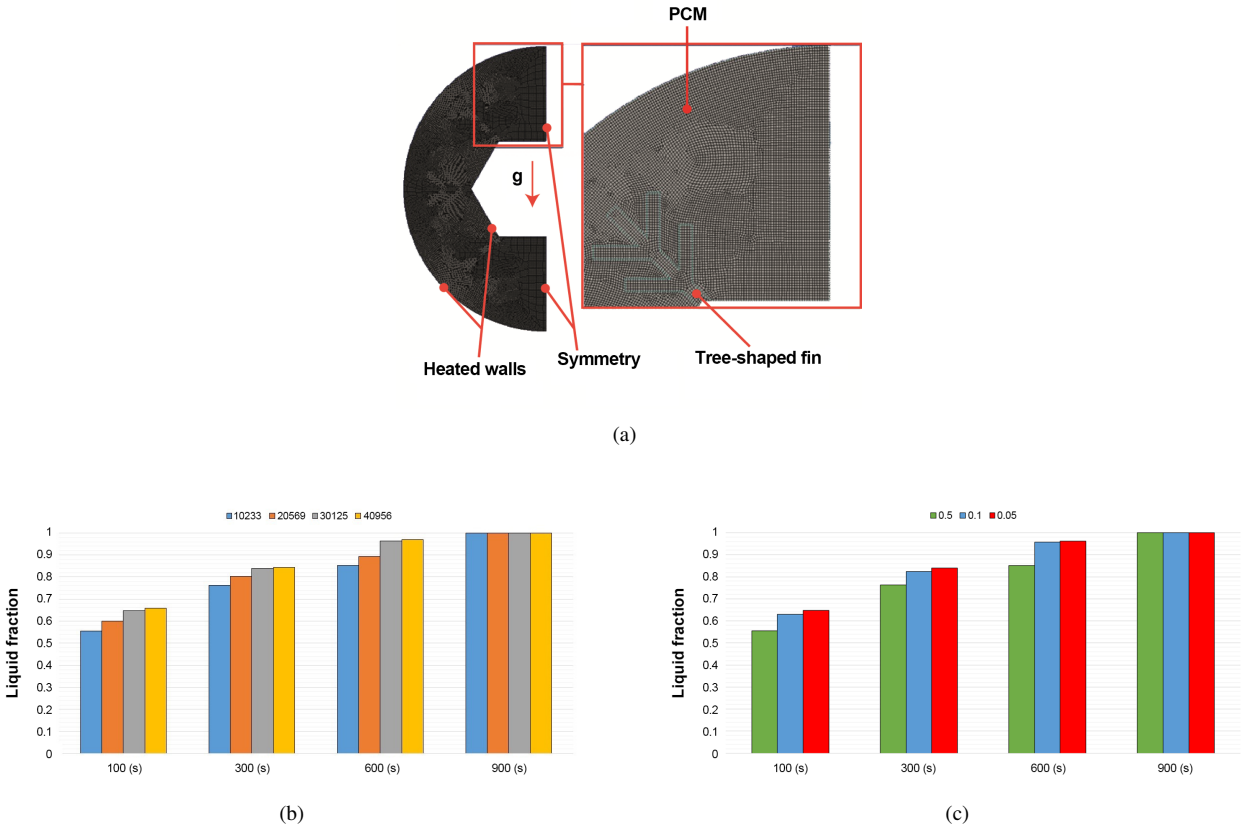


Figure 3. (a) Boundary conditions and mesh configuration; (b) grid independence analysis; (c) time-step independence analysis

Note: PCM = phase change material.

The overall cost of computation should be reduced without sacrificing the precision and dependability of the

numerical outcomes. The boundary conditions and a representative mesh arrangement for Case 2 are shown in Figure 3a. Figure 3b displays the relevant liquid fraction profiles for a range of grid sizes. Prior to the numerical simulation, a variety of grids should be evaluated in order to examine grid independence of the results. The resolution of the mesh does not significantly impact the values of the scalar once the number of grid elements exceeds a certain threshold. This number should be regarded as the ideal grid size for every case. To determine optimal mesh size, four grids with 10233, 20569, 30125, and 40956 elements were compared. Since further grid refinement does not significantly change the results, the final grid was chosen to consist of 30125 elements. The solution has not changed significantly with additional mesh size refinement. Melting is a transient process that requires an appropriate time step value. Small time step values should be taken into consideration due to the gravity effects. Figure 3c illustrates that a time-step size of 0.1 s is ideal. The numerical data should first be validated against previous studies to assess the reliability of the outcomes of the present study. The variation in output results is negligible, as seen in Figure 4. The validation of the current findings is confirmed by strong concordance between the present results and prior data [25]. As a result, the current numerical estimations are quite accurate. The melting and solidification times are influenced by the arrangement of tree-shaped fins, highlighting the need to carefully study the NEPCM behaviour with fin geometry. The geometry is symmetric, and the finite volume approach has been used to simulate half of the zone.

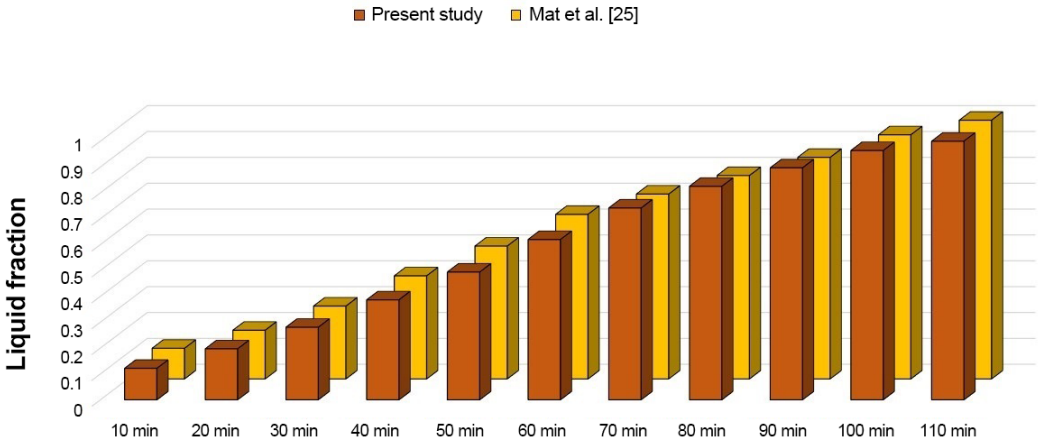


Figure 4. Validation of the present numerical model against the numerical findings reported by Mat et al. [25]

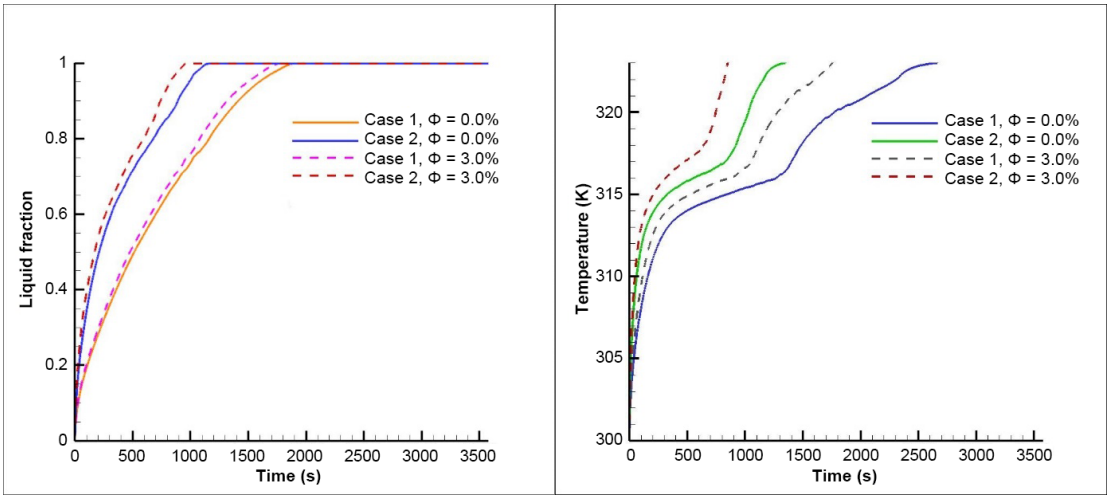


Figure 5. Transient profiles of liquid fraction and temperature in the heat exchanger system during the melting process for the examined cases

The temperature and liquid fraction changes throughout the charging process for different configurations at 0% and 3% nanoparticle concentrations are shown in Figure 5. The results show that increasing nanoparticle concentration diminishes melting duration and expedites the charging process. At volume fractions of 0% and 3%,

the melting completion durations were approximately 1860 s and 1701 s for Case 1 and 1052 s and 951 s for Case 2. Therefore, in Cases 1 and 2, the addition of nanoparticles reduced the melting time by 8.55% and 9.6%, respectively, whereas at 3% nanoparticle concentration, the tree-shaped fins decreased the melting time by 44.09%.

Figure 6 shows the time-dependent development of the liquid-fraction contours for the two configurations during the charging process at a 3% nanoparticle concentration. In Case 1, a thin layer of molten material forms around the heated hexagonal inner tube as a result of the charging phenomenon being mostly controlled by conduction. The lack of expanded fins limits heat penetration into the NEPCM domain and causes the liquid zone to expand gradually. For Case 2, tree-shaped fin arrangement offers a much larger effective heat transfer surface, which facilitates faster heat propagation within the NEPCM region. As charging advances, buoyancy-driven natural convection gains significance, causing the heated liquid NEPCM to rise, thereby enhancing fluid circulation inside the enclosure. At t = 600 s the higher conduction through the fins and natural convection in the molten NEPCM combine to give much larger liquid-fraction regions than in Case 1. In Case 1, the unmelted NEPCM is still available at t = 1200 s, while in Case 2, the melting process is terminated. This implies that the fin-induced enhancement of conductive pathways and the convection-driven heat transport can greatly accelerate phase change and improve overall thermal energy performance.

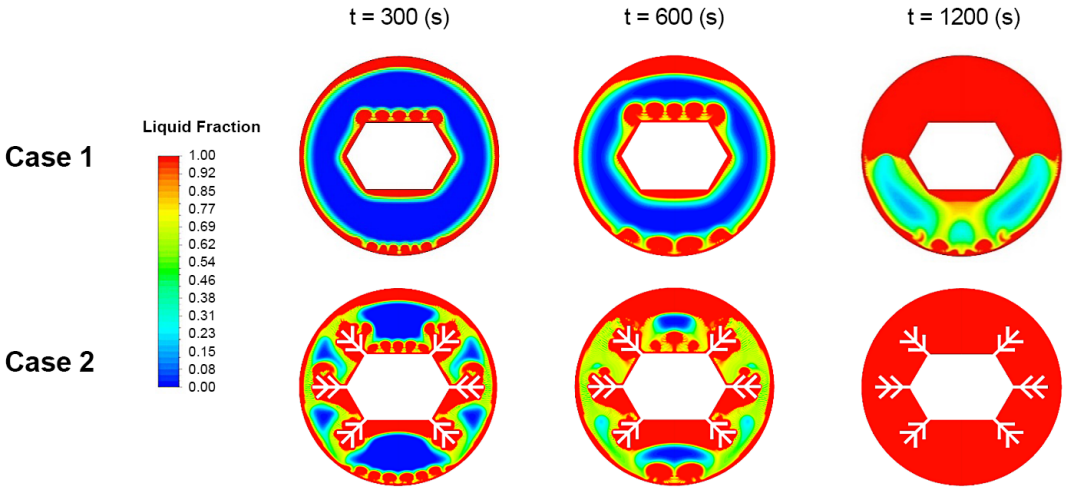


Figure 6. Transient development of liquid fraction contours throughout charging process for the studied cases

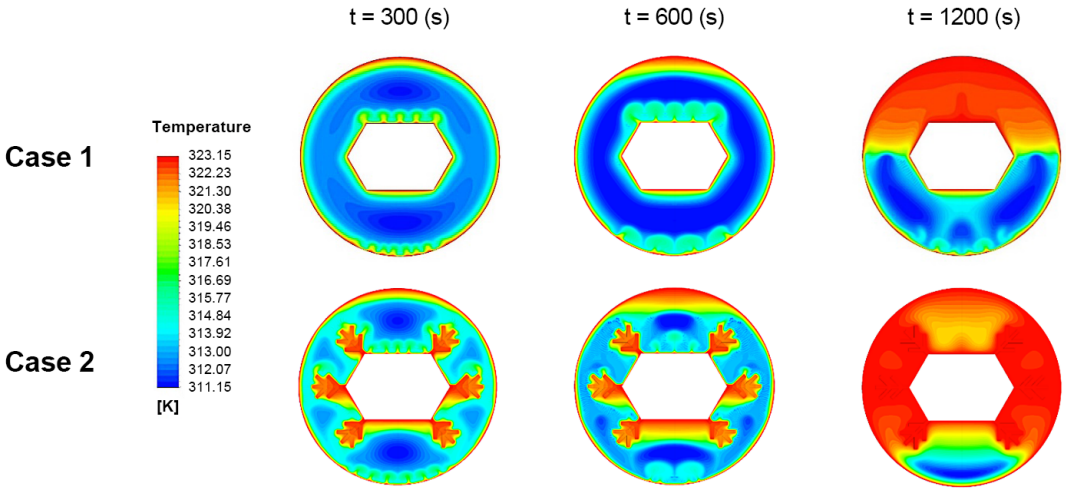


Figure 7. Temperature contours at different melting times for the investigated cases

The development of the temperature contours for the two configurations at 3% nanoparticle concentration during the melting process is shown in Figure 7. According to the temperature contours, Case 2 offers a more even heat distribution than Case 1. During early stages of melting, the high-temperature zone is confined around the heated hexagonal tube, showing that initially conduction dominates heat transfer and thermal penetration into the NEPCM

remains limited. The hexagonal structure creates thermal gradients at the corners, resulting in slightly enhanced heat transfer. Due to buoyancy-driven behavior of the molten NEPCM, the upper zone attains higher temperatures as time progresses. Conversely, Case 2 with tree-shaped fins facilitates heat transfer into the NEPCM domain, and extended high-temperature regions are formed around the fin surfaces. This improvement pathway reduces the cold zones and accelerates the development of the melting region. Consequently, the tree-shaped fin arrangement in Case 2 permits enhanced thermal homogeneity over the NEPCM region.

Figure 8 indicates the evolution of the velocity magnitude contours during the melting process for two configurations at nanoparticle concentrations of 3%. Since conduction primarily controls heat transfer in the early stages of melting, the velocity magnitude stays relatively low. Nevertheless, significant low-velocity zones are still observed, which can decrease the melting rate. As melting progresses, buoyancy forces are generated by temperature-induced density differences, resulting in natural convection currents that improve energy transport within the molten zone. Case 2 with tree-shaped fins creates multiple flow channels, leading to intensified fluid circulation and enhancing the local velocity around the fin branches. The buoyancy-driven convection and extended conductive pathways interaction leads to enhanced flow distribution compared with Case 1. At the end of the stage, Case 2 establishes stronger convective cells and facilitates heat diffusion throughout the NEPCM domain.

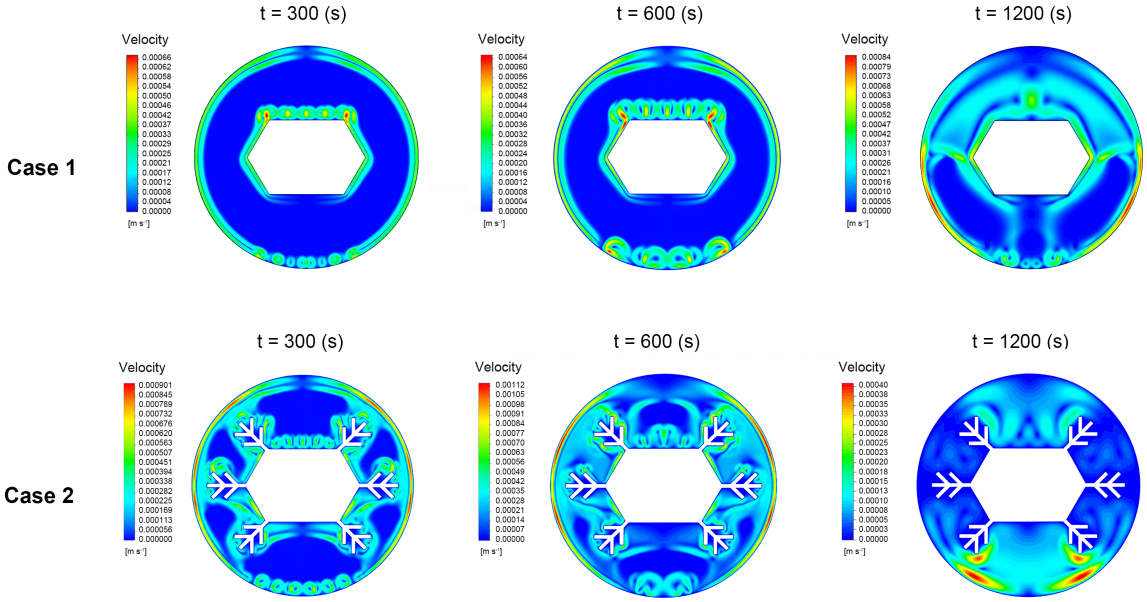


Figure 8. Evolution of velocity contours during the melting process for investigated cases

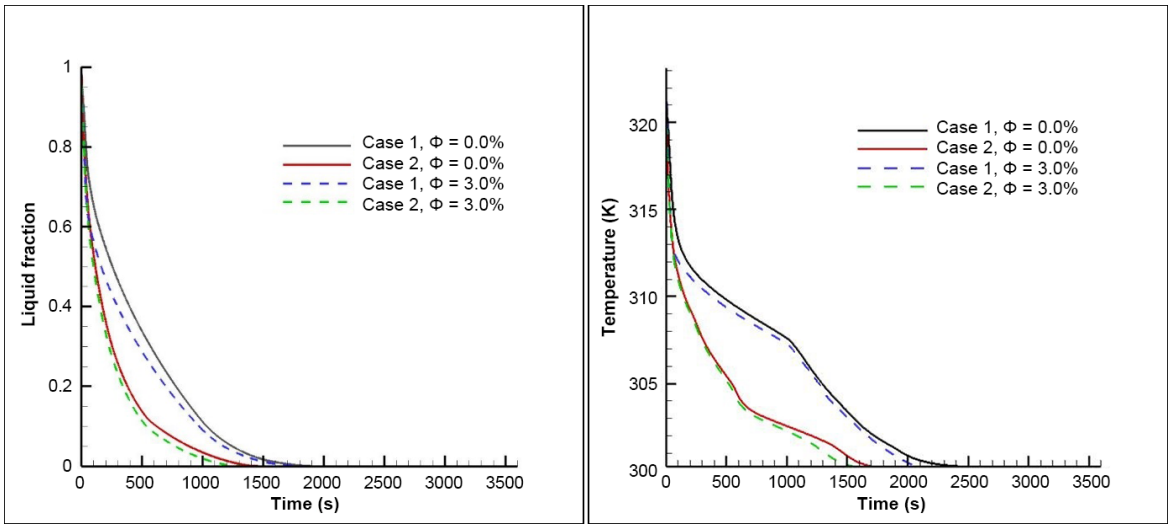


Figure 9. Transient profiles of liquid fraction and temperature during the solidification stage for the examined cases

Figure 9 displays liquid fraction and temperature variations for various cases at 0% and 3% nanoparticle concentrations throughout the discharging process. The findings demonstrate that a higher concentration of nanoparticles reduces discharging time and accelerates the solidification process. The solidification completion durations were around 1775 s and 1634 s for Case 1 and 1310 s and 1183 s for Case 2 at volume fractions of 0% and 3%, respectively. Thus, in Cases 1 and 2, the incorporation of nanoparticles reduced the solidification time by 7.94% and 9.69%, respectively, while the tree-shaped fins reduced the solidification time by 27.6% at a concentration of 3% nanoparticles.

The evolution of the liquid fraction for both configurations at a 3% nanoparticle concentration during the solidification process is shown in Figure 10. A significant quantity of liquid NEPCM is still present in the annular zone around the hexagonal tube in Case 1 during the early stages of solidification, indicating a sluggish rate of solidification. As solidification progresses, the liquid zone shrinks gradually; however, at t = 1200 s, some unfrozen NEPCM zones remain. Heat is conducted from the NEPCM more efficiently by tree-shaped fins, which accelerates the formation of solid zones around fin branches. Consequently, compared with Case 1, the remaining liquid NEPCM decreases more rapidly in Case 2, especially at t = 600 s and t = 1200 s. Overall, the tree-shaped fin structure enhances heat removal, promotes solidification performance, and reduces the extent of residual liquid regions.

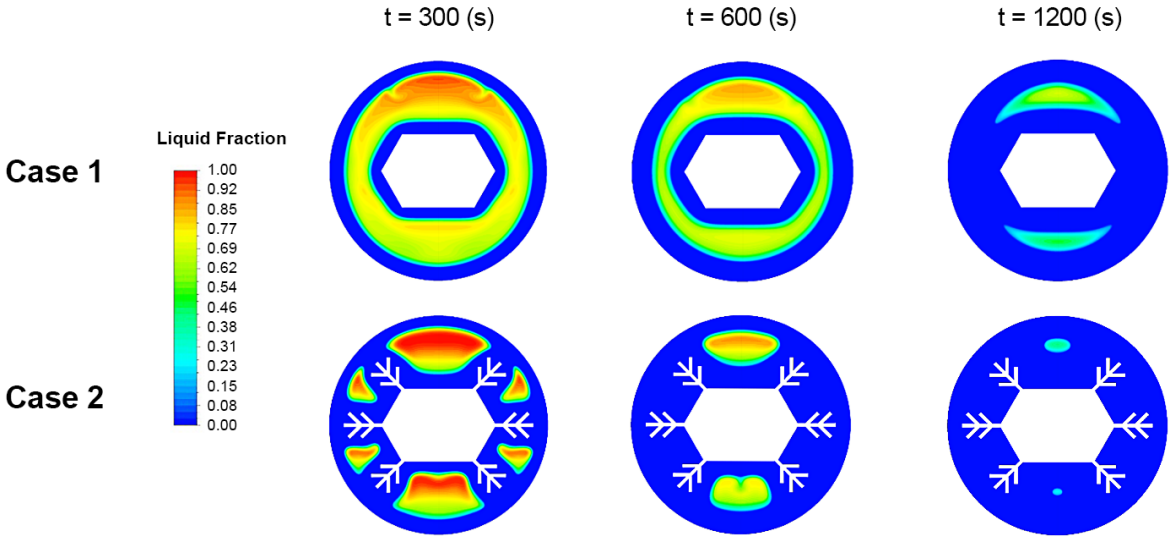


Figure 10. Transient development of liquid fraction contours throughout the solidification process for studied cases

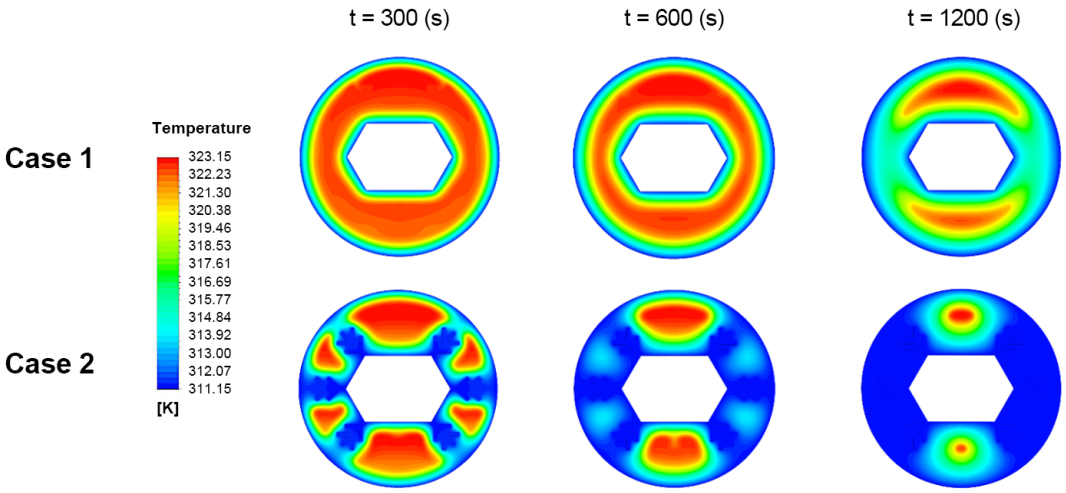


Figure 11. Transient evolution of temperature contours during the solidification process for studied cases

Figure 11 shows the time development of the temperature contours for the two cases at a 3% nanoparticle volume fraction during the discharging operation. The distributed heat-transfer interface in Case 1 is provided by the hexagonal inner tube, but the absence of tree-shaped fins limits heat penetration into the NEPCM domain, which

leads to large high-temperature regions that persist within the storage region. The corners of the hexagonal geometry increase local temperature gradients, while the branches of the fins propagate the cooling effect deeper into the NEPCM region. Consequently, the remaining elevated-temperature areas decrease more quickly as zones of lower temperature arise earlier. At the final stage, the high-temperature regions in Case 2 become confined to small isolated zones located far from the tree-shaped fin branches.

Figure 12 demonstrates the progression of velocity contours during the discharging process for the two configurations at nanoparticle concentrations of 3%. In Case 1, at the upper and lower regions noticeable velocity zones remain around the hexagonal tube. This behavior suggests buoyancy-driven motion is still active and the liquid NEPCM persists for a longer time. Due to the growth of the solidified layer, the velocity magnitude decreases as time increases. The fins in Case 2 divide the liquid zone into smaller flow channels. Consequently, the localized residual velocity zones replace the large circulation cells. At the final stage, only small localized velocity zones remain in Case 2, which indicates faster suppression of fluid motion. The use of the hexagonal tube and tree-shaped fins results in faster velocity decay, indicating enhanced heat extraction and earlier solidification.

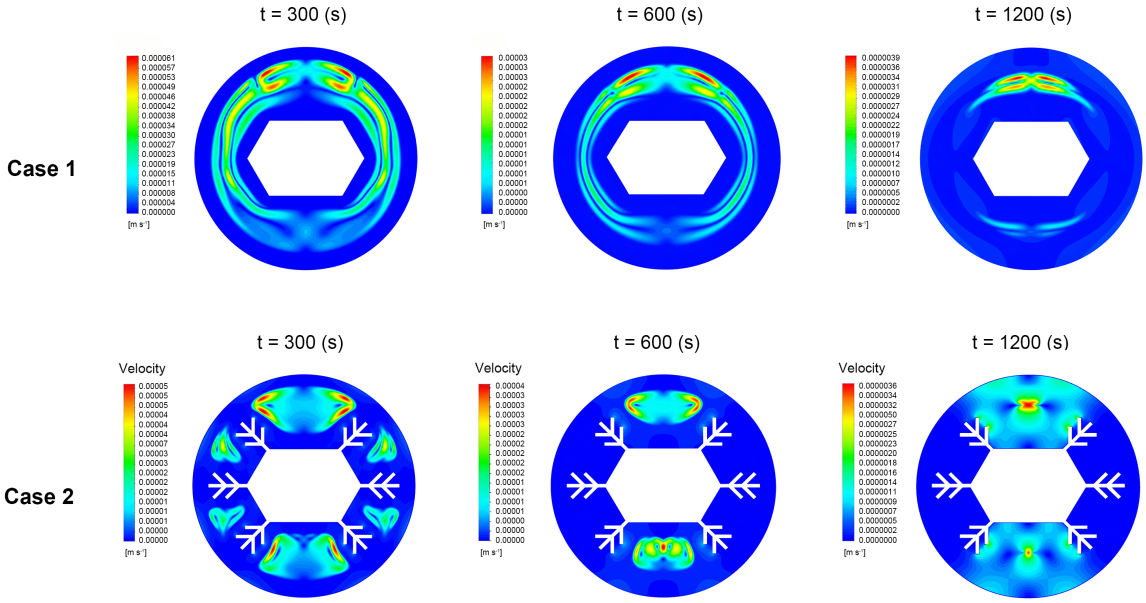


Figure 12. Velocity contours during the solidification process for the investigated cases

4 Conclusions

This study investigated effect of tree-shaped fins on enhancing charging and discharging rates in TES systems integrated with evacuated tube collectors. This type of heat exchanger has a wide range of industrial applications; in the current article, it was used as a container for the melting and solidification processes. The paraffin (RT42) was chosen due to its moderate melting point, making it suitable for applications involving moderate surface temperatures. In order to enhance the thermal conductivity coefficient, Al_2O_3 nanoparticles were incorporated into paraffin (RT42). The finite volume approach was used for the two-dimensional simulation of paraffin during melting and solidification processes. Two configurations employing the same volume of NEPCM were considered, with different thicknesses and fin arrangements. The equations governing laminar flow in the two-dimensional container included gravitational body forces and transient source terms. Due to the symmetric boundary condition, it was sufficient to analyze the system by considering half of the domain. The mathematical model was formulated using the enthalpy-porosity approach. Assuming that the liquid paraffin was an incompressible Newtonian fluid, the Boussinesq approximation was used to account for buoyancy effects arising from density variations in the gravity term. Both configurations contained the same volume of paraffin, and volume expansion was neglected. ANSYS Fluent was used to simulate the phase change model. Advection terms were discretised using second-order upwind, and transient terms were discretised using an implicit approach. The mesh resolution and time steps were examined for the design with tree-shaped fins at a nanoparticle volume fraction of 0.03 in order to lower the computational cost. Four grid resolutions were used to test grid independence, and three time-step sizes of 0.5 s, 0.1 s, and 0.05 s were examined for the time-step sensitivity analysis. The selected mesh with 30,125 elements and a time step of Δt = 0.1 s provided accurate results with the lowest computational cost. The suggested numerical approach was

verified by previous studies and results demonstrated that present outputs have good agreement with the literature. Additionally, velocity contours were presented to show the favorable effect of buoyancy forces in the liquid phase region. To extract contours for different configurations, three representative time points were selected. The heated temperature sustained by the inner and outer walls was much higher than the system's initial temperature. The results indicated that during both charging and discharging processes, the hexagonal tube geometry contributed effectively by increasing and creating the interaction between the heated/cooled walls and local thermal gradients near the tube corners. This structure facilitated thermal penetration from multiple angular regions around the inner tube during the melting process, while the utilization of tree-shaped fins created conductive channels extending deeper into the NEPCM region. Consequently, the hexagonal tube and tree-shaped fins improved temperature dispersion and accelerated liquid-phase development. In Cases 1 and 2, the addition of Al_2O_3 nanoparticles to PCM reduced the charging time by 8.55% and 9.6%, respectively. At a nanoparticle concentration of 3%, the tree-shaped finned arrangement reduced the melting time by 44.09% in comparison to the finless case. Moreover, the liquid fraction and velocity contours during the charging operation indicated that tree-shaped fins acted synergistically throughout the melting process. The tree-shaped fin branches divided the molten region into smaller local flow paths, while the hexagonal tube provided a more distributed heat-transfer boundary. This promoted stronger buoyancy-driven circulation and improved the melting rate throughout the NEPCM domain. The same geometric features enhanced heat removal from the NEPCM throughout the discharge operation. The tree-shaped fins propagated a cooling effect into the storage zone and accelerated the formation of solid layers around the fin branches, while the hexagonal tube increased local temperature gradients around its corners. The incorporation of Al_2O_3 nanoparticles shortened the solidification time by 7.94% and 9.69% in Cases 1 and 2, respectively, compared with the cases without nanoparticles. Additionally, at 3% nanoparticle concentration, the tree-shaped fin geometry decreased solidification time by 27.6% compared with the unfinned case. The velocity and temperature contours indicated that Case 2 decreased residual high-temperature zones and caused an earlier decay of buoyancy-driven motion. The tree-shaped fin geometry was essential for increasing conductive pathways and improving local flow development, while the hexagonal tube improved the distribution of heat transfer during charging and discharging processes.

Author Contributions

Conceptualization, M.J.S.; software, M.R., M.R.D., and M.J.S.; formal analysis, M.R.D. and M.J.S.; investigation, M.J.S.; resources, M.R.D.; data curation, M.R.D.; writing—original draft preparation, M.R.; writing—review and editing, M.R.; supervision, M.R. All authors have read and agreed to the published version of the manuscript.

Data Availability

Not applicable.

Conflicts of Interest

The authors declare no conflicts of interest.

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